

Vim Game

Deployment Manual

Project: Vim Game
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1. Overview

Vim Game is a browser-based Angular application with Firebase as its backend. There is no custom server — the frontend is served as a static build, and all backend functionality (authentication, data storage) is handled by Firebase.

Layer	Technology
Frontend	Angular 21
Auth & Database	Firebase (Authentication + Firestore)
Hosting	Firebase Hosting (recommended) or any static host
Language	TypeScript 5.9

2. Dependencies

Runtime Dependencies

Package	Version	Purpose
@angular/common	^21.2.0	Angular core modules
@angular/router	^21.2.0	Client-side routing
@angular/material	^21.2.10	UI components
firebase	^12.12.1	Authentication & data storage

Dev Dependencies

Package	Version	Purpose
@angular/cli	^21.2.3	Build & serve tooling
@angular/build	^21.2.3	esbuild-based bundler
typescript	~5.9.2	TypeScript compiler
vitest	^4.0.8	Unit testing

External Services

Service	Purpose	Required
Firebase Authentication	User login & registration	Yes
Firebase Firestore	Storing user progress	Yes
Google Fonts	Press Start 2P pixel font	No

3. Prerequisites

The following must be installed on the deployment machine:

Tool	Minimum Version	Install
Node.js	18	nodejs.org
npm	10.8.2	Included with Node.js
Angular CLI	21	npm install -g @angular/cli

Verify installations:

```
node --version npm --version ng version
```

4. Environment Configuration

The application requires a Firebase configuration file that is not included in the repository (listed in .gitignore for security).

Create the file `src/environments/environment.ts` with the following structure:

```
export const environment = { production: false, skipAuthGuard: false, firebase: { apiKey: 'YOUR_API_KEY', authDomain: 'YOUR_PROJECT.firebaseio.com', projectId: 'YOUR_PROJECT_ID', storageBucket: 'YOUR_PROJECT.firebaseiostorage.app', messagingSenderId: 'YOUR_SENDER_ID', appId: 'YOUR_APP_ID', }, };
```

The actual values can be found in the Firebase Console under Project Settings > Your Apps.

■■ *Never commit environment.ts to the repository. It contains sensitive API keys.*

5. Local Deployment

```
# 1. Clone the repository git clone <repository-url> cd vim-game # 2. Install dependencies
npm install # 3. Create environment file (see Section 4) # 4. Start the development server
ng serve
```

The application is available at `http://localhost:4200`.

6. Production Build

To build an optimized production bundle:

```
ng build --configuration production
```

The output is generated in the `dist/` folder. This folder contains all static files (HTML, CSS, JS) that can be served by any static web host.

Set production: true and skipAuthGuard: false in environment.ts before building.

7. Firebase Deployment

Firebase Hosting is the recommended deployment target as the project already uses Firebase for auth and data.

Setup (first time only)

```
# Install Firebase CLI npm install -g firebase-tools # Login firebase login # Initialize
hosting firebase init hosting
```

During firebase init:

- Select the existing Firebase project (vim-game-dev)
- Set public directory to dist/vim-game/browser
- Configure as single-page app: Yes
- Do not overwrite index.html: No

Deploy

```
ng build --configuration production firebase deploy --only hosting
```

After deployment, Firebase provides a live URL: <https://vim-game-dev.web.app>